Care Transition Efficiency & Placement Outcome Analytics

Abstract

The Unaccompanied Alien Children (UAC) Program operates as a structured care and reunification pipeline rather than a conventional healthcare system. Children move through multiple stages including apprehension, CBP custody, transfer to HHS care, medical screening, sheltering, case management, and eventual reunification with a vetted sponsor. While official reports often track the total number of children in custody, limited attention is given to how efficiently children move through these stages.

This research focuses on evaluating the process efficiency, delays, and placement outcomes within the UAC care pipeline. Using transition-based metrics and time-series analysis, the study identifies system bottlenecks, measures discharge effectiveness, and analyzes backlog accumulation. An interactive Streamlit dashboard is developed to visualize these metrics and support data-driven decision-making. The findings highlight the importance of process efficiency in improving reunification timelines and strengthening child welfare outcomes.

Keywords: UAC Program, Care Pipeline, Process Efficiency, Bottleneck Analysis, Placement Outcomes, Streamlit Dashboard

1 Introduction

The Unaccompanied Alien Children (UAC) Program, managed by the U.S. Department of Health and Human Services, is designed to ensure the safe care and timely reunification of children who enter the United States without a parent or legal guardian. The program functions as a multi-stage pipeline that begins with apprehension and CBP custody and continues through HHS care, medical screening, sheltering, case management, and discharge to a vetted sponsor.

From a humanitarian and policy perspective, the effectiveness of this system depends not only on capacity but also on how smoothly and quickly children move through each stage. Delays in any part of the pipeline can result in longer stays in custody, increased operational stress, and negative outcomes for children. However, most monitoring practices emphasize total counts rather than system efficiency.

This project shifts the analytical focus from capacity monitoring to process efficiency

and outcome evaluation, enabling a clearer understanding of how well the UAC care pipeline functions in practice.

2 Problem Statement

Although aggregate statistics on children in CBP and HHS custody are publicly available, several important operational questions remain unanswered:

- How efficiently are children transferred from CBP custody to HHS care?
- Are discharge and sponsor placement rates keeping pace with daily inflows?
- When and where do care backlogs accumulate?
- Are placement outcomes stable, improving, or deteriorating over time?

Without structured transition analytics, bottlenecks and inefficiencies remain difficult to detect, limiting the ability of administrators and policymakers to improve system performance.

3 Objectives

Primary Objectives

- To measure the efficiency of transitions from CBP custody to HHS care
- To evaluate discharge and sponsor placement outcomes
- To identify delays and bottlenecks within the care pipeline

Secondary Objectives

- To support faster reunification timelines
- To improve case management workflows
- To provide data-driven insights for policy-level reforms

4 Dataset Description

The analysis uses a daily time-series dataset published under the UAC Program. The dataset captures inflows, active care loads, and successful exits across different stages of the pipeline.

Column	Description
Date	Daily reporting date
Children apprehended and placed in CBP custody	Daily intake volume
Children in CBP custody	Active CBP care load
Children transferred out of CBP custody	Transfers into HHS care
Children in HHS Care	Active HHS care load
Children discharged from HHS Care	Successful sponsor placements

5 Analytical Methodology

5.1 Care Pipeline Modeling

The UAC system is modeled as a three-stage flow pipeline:

$$\text{CBP Custody} \rightarrow \text{HHS Care} \rightarrow \text{Sponsor Placement}$$

Daily movements between these stages are tracked to analyze how efficiently children progress through the system and where delays occur.

5.2 Transition Efficiency Metrics

To evaluate system performance, the following metrics are calculated:

- **Transfer Efficiency Ratio:** Measures how effectively children are transferred from CBP custody to HHS care.
- **Discharge Effectiveness Index:** Measures the success rate of sponsor placements from HHS care.
- **Pipeline Throughput Rate:** Compares total exits to total entries to assess overall system movement.

5.3 Backlog and Delay Identification

Backlogs are identified by comparing daily inflows with successful discharges. When inflows exceed exits for extended periods, unresolved cases accumulate, indicating system stress and operational delays.

5.4 Temporal Analysis

Time-series analysis is used to observe long-term trends and seasonal patterns, helping identify periods of stagnation and changes in placement performance over time.

5.5 Outcome Stability Analysis

Outcome stability is evaluated by analyzing variability in discharge effectiveness. Sudden fluctuations or prolonged instability highlight weaknesses in sponsor placement processes.

6 Key Performance Indicators (KPIs)

KPI	Description
Transfer Efficiency Ratio	Speed of CBP to HHS transitions
Discharge Effectiveness Index	Sponsor placement success
Pipeline Throughput	Overall system flow
Backlog Accumulation Rate	Severity of delays
Outcome Stability Score	Consistency of placements

7 Streamlit Dashboard Implementation

To operationalize the analysis, an interactive Streamlit dashboard was developed and deployed on Streamlit Community Cloud.

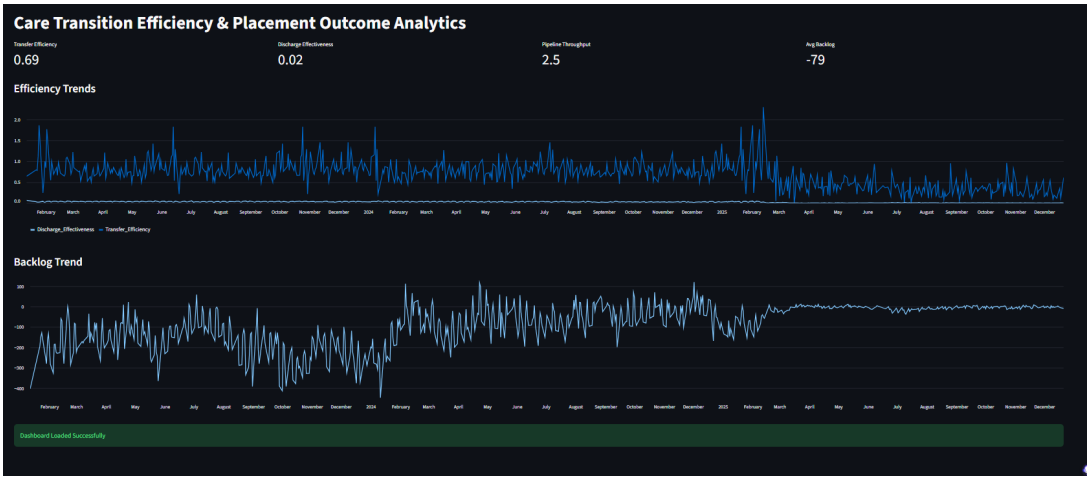

Figure 1: Streamlit dashboard showing key performance indicators for care transition efficiency and placement outcomes.

The dashboard provides real-time visibility into efficiency trends and backlog patterns through interactive filters and visualizations.

8 Results and Insights

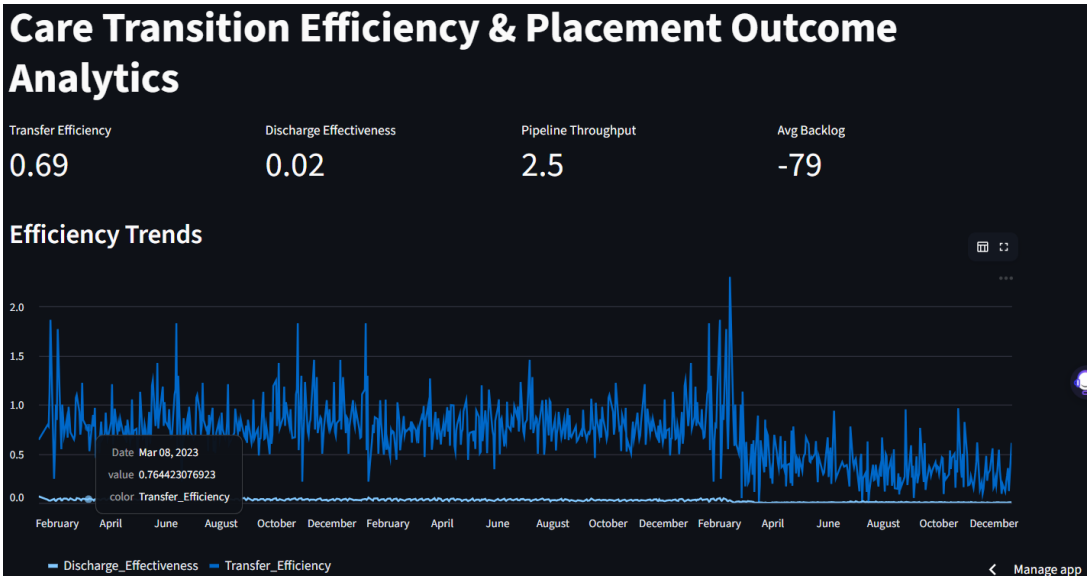

Figure 2: Time-series trends of CBP to HHS transfer efficiency and discharge effectiveness.

Transfer efficiency remains relatively stable but shows fluctuations during high intake periods, indicating congestion at the CBP stage. Discharge effectiveness is consistently lower, suggesting sponsor placement as the primary bottleneck.

Figure 3: Backlog accumulation trend highlighting periods of imbalance between inflows and successful discharges.

Backlog accumulation increases when discharges fail to keep pace with inflows, leading to sustained system stress.

9 Policy and Operational Recommendations

- Improve coordination and capacity for CBP to HHS transfers during surge periods
- Streamline sponsor vetting and placement procedures
- Use backlog and efficiency KPIs as early warning indicators
- Introduce flexible staffing and processing during high-demand periods
- Continuously monitor outcome stability to prevent system overload

10 Conclusion

This project reframes the UAC dataset from a basic reporting tool into a process efficiency and outcome evaluation framework. By focusing on how children move through the care pipeline, the analysis identifies hidden bottlenecks, evaluates placement performance, and provides actionable insights for improving reunification timelines. The integration of analytical modeling with an interactive Streamlit dashboard supports continuous monitoring and evidence-based decision-making, ultimately contributing to stronger child welfare outcomes.

References

- U.S. Department of Health and Human Services – UAC Program Data
- Operations and Process Analytics Literature
- Streamlit Documentation